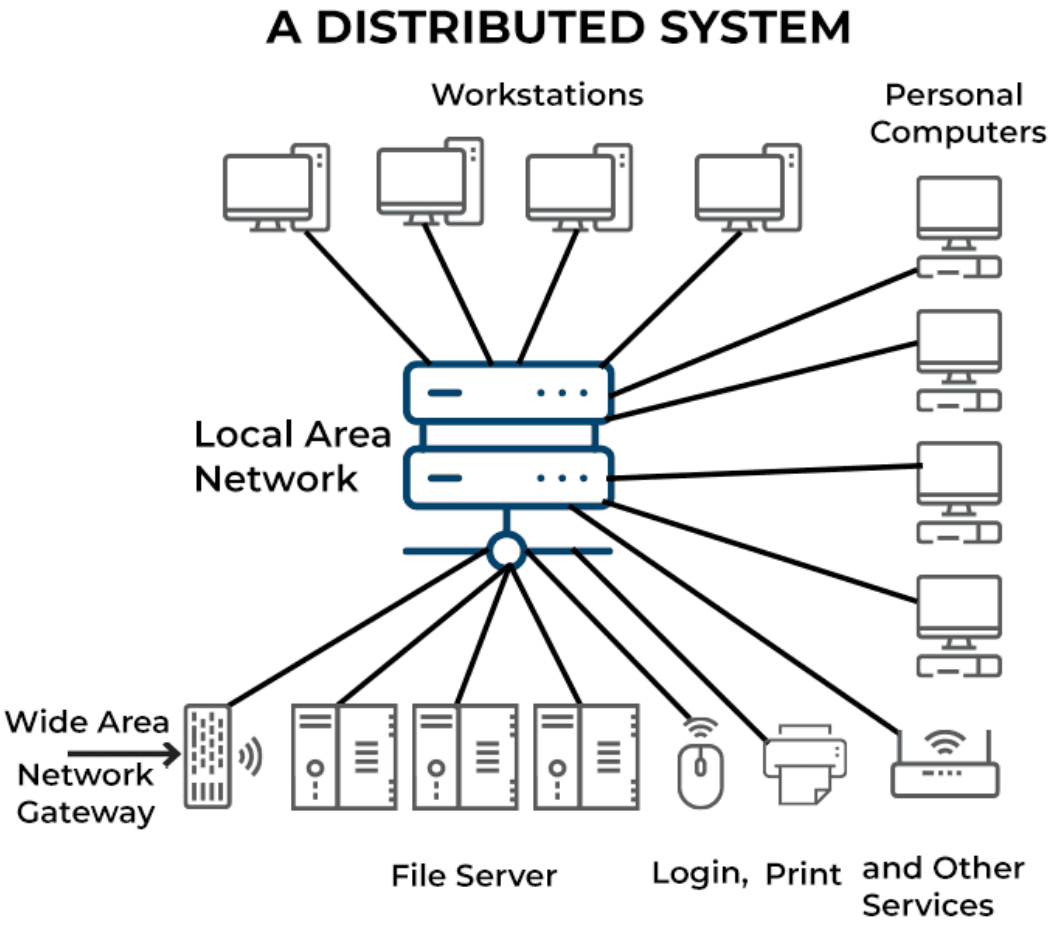
CS 432 Distributed Systems	Lab (1)

Section Content:

- **Introduction to Distributed Systems**

What is a Distributed System?

- **Definition 1:** A software system in which components located on networked computers **communicate and coordinate** their actions only by passing messages.
- **Definition 2:** A collection of independent computers that appears to its users as a **single coherent system**.
- **Definition 3:** A multi-machine system designed **to process massive data**.



Why do we need Distributed Systems?

Because it is to solve these two problems

1. **Performance Limitations:** A single machine is slow when handling huge tasks.
2. **Storage Capacity Limitations:** A single machine can't hold all the data.

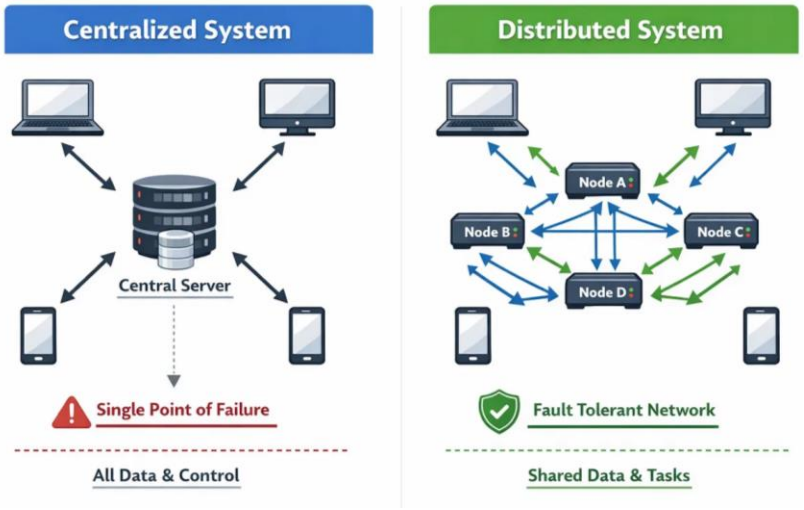
Two basic core types of distributed systems

1. **Distributed Computing Framework:** To solve the problem of slow calculation.
2. **Distributed Storage Engine:** To solve the problem of storing massive data.

Key Characteristics of a Distributed System

1. **Scalability:** The ability to grow as the size of the workload increases is an essential feature of distributed systems, accomplished by adding additional processing units or nodes to the network as needed.
2. **Concurrency:** Distributed system components run simultaneously. They're also characterized by the lack of a "global clock," when tasks occur out of sequence and at different rates.
3. **Availability and fault tolerance:** If one node fails, the remaining nodes can continue to operate without disrupting the overall computation.
4. **Heterogeneity:** In most distributed systems, the nodes and components are often asynchronous, with different hardware, middleware, software and operating systems.
5. **Replication:** Distributed systems enable shared information and messaging, ensuring consistency between redundant resources, such as software or hardware components, improving fault tolerance, reliability and accessibility.
6. **Transparency:** The end user sees a distributed system as a single computational unit rather than as its underlying parts, allowing users to interact with a single logical device rather than being concerned with the system's architecture.

Why distributed systems is better than centralized systems?



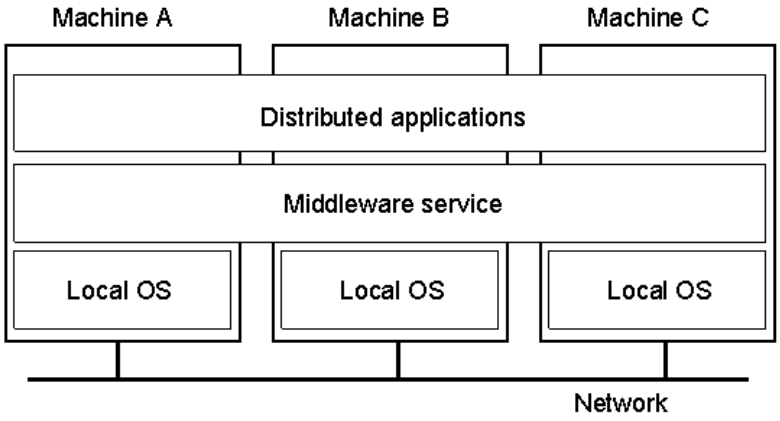
There are three reasons

1. **Low latency:** Distributed systems have **low** latency because of high geographical spread, hence leading to **less** time to get a response.
2. **Scalability:** Distributed systems are highly scalable as they can be **easily expanded** by adding new nodes to the network.
3. **Better Performance:** By using the resources of numerous computers to tackle the workload, distributed systems can perform at higher levels than centralized systems.

Distributed Systems Layout

Distributed system?

1. Autonomous systems
2. Computer network
3. Distributed software



Important consequences:

- Concurrency
- No global clock
- Independent failure

Middleware Architecture

Middleware acts as a bridge. It refers to software that sits between different software applications or systems to facilitate communication, data management, and interaction between them. It provides a set of services and functionalities to higher-level applications.

A software **layer** that masks the heterogeneity of systems

Advantages & disadvantages distributed systems

Advantages

- interaction
- co-operation
- sharing of resources
- reduced costs
- improved availability and performance
- Scalability
- fault tolerance

Disadvantages

- **Complexity:** Typically, distributed systems are more complex than centralized systems
- **Security:** More susceptible to external attack
- **Manageability:** More effort required for system management
- **Unpredictability:** Unpredictable responses depending on the system organization and network load

Typical distributed systems Examples

1. **The Internet**
   A global network of interconnected computers which communicate through IP Protocols.
   - **IP Address:** Unique address that allows devices to connect on a network.
   - **Protocol:** A set of rules providing specific services for the OS (e.g., HTTP for web content).

   Characteristics of the Internet:
   1. **Very Large & Heterogeneous:** Combines vast numbers of different hardware and software types.
   2. **Multiple Services Enabled:** Supports email, file transfer, multimedia communications, WWW.
   3. **Open-Ended:** Has no boundaries; continuously expandable.
2. **Intranet (Intra = Within)**
   A separately administered network with boundaries. It is a private network used within companies, schools, or government agencies to facilitate internal communication and resource sharing among employees or members.
   - **Connectivity:** Connected to the Internet via Routers.
   - **Security:** In/Out communications are protected by a Firewall.
   - **Examples of Organizations Using Intranets:** French University in Egypt, Egypt Post, Egypt's Ministry of Petroleum Intranet.
3. **Extranet (Extra = Outside/Extended)**
   An organization's intranet that is extended to authorized users outside the organization.
4. **Facebook (Meta)**
   Facebook connects billions of users worldwide and relies on a set of custom-built distributed systems to store and serve the social graph, photos, and messages at massive scale.
5. **Amazon**
   Amazon uses distributed systems both to run its e-commerce platform and to offer distributed computing as a service through AWS (Amazon Web Services).

6. Google

Google pioneered several of the foundational technologies that modern distributed systems are built on, in order to index and search the entire web at massive scale.